# Data Analysis with Python Demo

REX_SYNTHETIC_DEMO | Fictional TA example. No real academy offering, instructor, accreditation or tender.

Course ID: DEMO-DA-201

Title: Data Analysis with Python Demo

Description: A fictional applied course in cleaning tabular data, calculating business metrics and communicating findings with Python and pandas. All exercises use invented sales data.

Level: Intermediate

Training format: Hybrid

Price EGP: 4800 EGP

Duration hours: 36

Department: Data and AI

Instructor name: Salma Demo Instructor

Instructor bio: Fictional analyst instructor for demonstration only.

Schedule: Demo cohort: 12 sessions of 3 hours, Monday and Wednesday 18:00-21:00 Cairo time; dates unconfirmed.

Offers: No discount is specified for this fictional course.

Target audience: Learners comfortable with Python functions and CSV files.

Career opportunities: Foundation for junior data analysis; Reporting and operations analytics

Admission requirements: Laptop with Python and pandas; Complete a basic Python exercise

Prerequisites: Python variables, loops and functions; Basic spreadsheet arithmetic

Learning outcomes: Clean missing and duplicate values; Calculate grouped sales metrics; Explain a chart without making causal claims

FAQs: [{"question": "Can a complete beginner join?", "answer": "Python foundations are required before this intermediate example."}]

## Course content

This material is an authored fixture for testing ingestion and document workflows. It is not an accredited curriculum.

## Inspect and clean

Read sales.csv with pandas. Check column types and missing values. Remove exact duplicate rows and document why an absent quantity must be checked before filling.

## Calculate metrics

Calculate revenue as quantity multiplied by unit_price. Group revenue by category. Verify two units at 150 produce 300 and never sum unit prices as if they were revenue.

## Visualize and communicate

Create a bar chart by category with labeled axes and currency. Explain sample size and missing data. State that a correlation does not demonstrate a causal relationship.

## Analysis project

Submit a reproducible notebook with input checks, cleaned output, three findings and one limitation. Assess data quality 35 percent, calculations 35 percent and communication 30 percent.